

Exercise 1

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Plotly is a plotting library with bindings for different programming languages. It enables the creation of interactive plots based on HTML, CSS and javascript. An introduction to the parallel coordinates plot with plotly can be found here:

<https://plotly.com/python/parallel-coordinates-plot/> In this exercises, we will work with a Mice Protein Expression Dataset, which contains expression levels of 77 proteins, measured in the cerebral cortex of 8 classes of mice. The classes result from two genotypes (Ts65Dn, which serves as a mouse model of human down syndrome, vs. normal controls), two treatments (injection of the drug memantine vs. a saline solution as a control), and two experimental conditions related to context fear conditioning (context-shock, which should lead to learning, vs. shock-context, in which no learning takes place). Counting all repeated measurements, there are 1080 samples overall, some with missing data. You can find more information on the data in the corresponding publication.

```
In [1]: #We begin by installing plotly  
!mamba install plotly  
!mamba install pandas  
!mamba install xlrd  
!mamba install nbformat
```

mambajs 0.19.13

Specs: xeus-python, numpy, matplotlib, pillow, ipywidgets>=8.1.6, ipyleaflet, sci
py, plotly

Channels: emscripten-forge, conda-forge

Solving environment...

Solving took 0.943 seconds

Name Channel	Version	Build
+ narwhals conda-forge	2.20.0	pyhcf101f3_0
+ plotly conda-forge	6.6.0	pyhd8ed1ab_0
- pip conda-forge	25.3	pyh145f28c_0
mambajs 0.19.13		

Specs: xeus-python, numpy, matplotlib, pillow, ipywidgets>=8.1.6, ipyleaflet, sci
py, plotly, pandas

Channels: emscripten-forge, conda-forge

Solving environment...

Solving took 0.38730000001192094 seconds

Name Channel	Version	Build
+ pandas emscripten-forge	3.0.2	np22py313h9d9dc1e_0
+ python-tzdata conda-forge	2026.1	pyhd8ed1ab_0
mambajs 0.19.13		

Specs: xeus-python, numpy, matplotlib, pillow, ipywidgets>=8.1.6, ipyleaflet, sci
py, plotly, pandas, xlrd

Channels: emscripten-forge, conda-forge

Solving environment...

Solving took 0.3480999999940395 seconds

Name Channel	Version	Build
+ xlrd conda-forge	2.0.2	pyhd8ed1ab_0
mambajs 0.19.13		

Specs: xeus-python, numpy, matplotlib, pillow, ipywidgets>=8.1.6, ipyleaflet, sci
py, plotly, pandas, xlrd, nbformat

Channels: emscripten-forge, conda-forge

Solving environment...

Solving took 0.32310000002384187 seconds

Name Channel	Version	Build

+ attrs	26.1.0	pyhcf101f3_0
conda-forge		
+ importlib_resources	7.1.0	pyhd8ed1ab_0
conda-forge		
+ importlib-metadata	8.8.0	pyhcf101f3_0
conda-forge		
+ jsonschema	4.17.3	pyhd8ed1ab_0
conda-forge		
+ jupyter_core	5.9.1	py313h1804a44_1
emscripten-forge		
+ nbformat	5.10.4	pyhd8ed1ab_1
conda-forge		
+ pkgutil-resolve-name	1.3.10	pyhd8ed1ab_2
conda-forge		
+ platformdirs	4.9.6	pyhcf101f3_0
conda-forge		
+ pyrsistent	0.20.0	py313hd355c7d_2
emscripten-forge		
+ python-fastjsonschema	2.21.2	pyhe01879c_0
conda-forge		
+ zipp	3.23.1	pyhcf101f3_0
conda-forge		

```
In [2]: import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import xlrd
import plotly.express as px
```

```
In [48]: !mamba list nbformat
```

Name Channel	Version	Build
asttokens	3.0.1	pyhd8ed1ab_0
conda-forge		
attrs	26.1.0	pyhcf101f3_0
conda-forge		
backcall	0.2.0	pyh9f0ad1d_0
conda-forge		
branca	0.8.2	pyhd8ed1ab_0
conda-forge		
comm	0.2.3	pyhe01879c_0
conda-forge		
contourpy	1.3.3	py313h3a67976_1
emscripten-forge		
cycler	0.12.1	pyhcf101f3_2
conda-forge		
decorator	5.2.1	pyhd8ed1ab_0
conda-forge		
emscripten-abi	3.1.73	h267e887_12
emscripten-forge		
executing	2.2.1	pyhd8ed1ab_0
conda-forge		
fonttools	4.39.4	py313h6e18b15_1
emscripten-forge		
importlib_resources	7.1.0	pyhd8ed1ab_0
conda-forge		
importlib-metadata	8.8.0	pyhcf101f3_0
conda-forge		
ipyleaflet	0.20.0	pyhd8ed1ab_0
conda-forge		
ipython	9.9.0	py313hd355c7d_0
emscripten-forge		
ipywidgets	8.1.8	pyhd8ed1ab_0
conda-forge		
jinja2	3.1.6	pyhcf101f3_1
conda-forge		
jsonschema	4.17.3	pyhd8ed1ab_0
conda-forge		
jupyter_core	5.9.1	py313h1804a44_1
emscripten-forge		
jupyter_leaflet	0.20.0	pyhd8ed1ab_0
conda-forge		
jupyterlab_widgets	3.0.16	pyhcf101f3_1
conda-forge		
kiwisolver	1.4.9	py313ha0aad71_1
emscripten-forge		
markupsafe	3.0.3	py313hd355c7d_0
emscripten-forge		
matplotlib	3.10.6	py313hb94a407_1
emscripten-forge		
matplotlib-base	3.10.6	np22py313h4202e73_1
emscripten-forge		
matplotlib-inline	0.2.1	pyhd8ed1ab_0
conda-forge		
narwhals	2.20.0	pyhcf101f3_0
conda-forge		
nbformat	5.10.4	pyhd8ed1ab_1
conda-forge		

nlohmann_json-abi	3.12.0	h0f90c79_1
conda-forge		
numpy	2.4.1	py313h6394566_0
emscripten-forge		
openblas-flang	0.3.30	h4e94343_3
emscripten-forge		
packaging	25.0	pyh29332c3_1
conda-forge		
pandas	3.0.2	np22py313h9d9dc1e_0
emscripten-forge		
pexpect	4.9.0	pyhd8ed1ab_1
conda-forge		
pickleshare	0.7.5	pyhd8ed1ab_1004
conda-forge		
pillow	12.1.0	py313hafa190e_0
emscripten-forge		
pkgutil-resolve-name	1.3.10	pyhd8ed1ab_2
conda-forge		
platformdirs	4.9.6	pyhcf101f3_0
conda-forge		
plotly	6.6.0	pyhd8ed1ab_0
conda-forge		
prompt-toolkit	3.0.52	pyha770c72_0
conda-forge		
ptyprocess	0.7.0	pyhd8ed1ab_1
conda-forge		
pure_eval	0.2.3	pyhd8ed1ab_1
conda-forge		
pygments	2.19.2	pyhd8ed1ab_0
conda-forge		
pyjs-rt	3.2.0	py313h7595f35_0
emscripten-forge		
pyodide-http	0.2.2	pyhcf101f3_0
conda-forge		
pyparsing	3.3.1	pyhcf101f3_0
conda-forge		
pyrsistent	0.20.0	py313hd355c7d_2
emscripten-forge		
python	3.13.1	h_c8de616_5_cp313
emscripten-forge		
python_abi	3.13.1	1_cp313
emscripten-forge		
python-dateutil	2.9.0.post0	pyhe01879c_2
conda-forge		
python-fastjsonschema	2.21.2	pyhe01879c_0
conda-forge		
python-tzdata	2026.1	pyhd8ed1ab_0
conda-forge		
pytz	2025.2	pyhd8ed1ab_0
conda-forge		
qhull	2020.2	h7223423_0
emscripten-forge		
scipy	1.17.0	np22py313h759dc5e_0
emscripten-forge		
six	1.17.0	pyhe01879c_1
conda-forge		
stack_data	0.6.3	pyhd8ed1ab_1
conda-forge		
traitlets	5.14.3	pyhd8ed1ab_1
conda-forge		

traitletypes	0.2.3	pyh332efcf_0
conda-forge		
typing_extensions	4.15.0	pyhcf101f3_0
conda-forge		
wcwidth	0.2.14	pyhd8ed1ab_0
conda-forge		
widgetsnbextension	4.0.15	pyhd8ed1ab_0
conda-forge		
xeus	5.2.6	h2072262_0
emscripten-forge		
xeus-python	0.17.6	py313h027658c_3
emscripten-forge		
xeus-python-shell	0.6.6	pyh332efcf_0
conda-forge		
xeus-python-shell-lite	0.6.6	pyhd8ed1ab_0
conda-forge		
xeus-python-shell-raw	0.6.6	pyhd8ed1ab_0
conda-forge		
xlrd	2.0.2	pyhd8ed1ab_0
conda-forge		
xyzservices	2025.11.0	pyhd8ed1ab_0
conda-forge		
zipp	3.23.1	pyhcf101f3_0
conda-forge		
zlib	1.3.1	h4e94343_2
emscripten-forge		

a) Use pandas to read the file *Data_Cortex_Nuclear.xls* provided via eCampus. Extract subgroups t-CS-s and c-CS-s. How many mice were measured for each class? (1P)

```
In [3]: data = pd.read_excel("Data_Cortex_Nuclear.xls", index_col = 0)
data.head()
```

```
Out[3]:
```

	DYRK1A_N	ITSN1_N	BDNF_N	NR1_N	NR2A_N	pAKT_N	pBRAf_N	pCAI
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MouseID

309_1	0.503644	0.747193	0.430175	2.816329	5.990152	0.218830	0.177565	2.
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309_2	0.514617	0.689064	0.411770	2.789514	5.685038	0.211636	0.172817	2.
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309_3	0.509183	0.730247	0.418309	2.687201	5.622059	0.209011	0.175722	2.
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309_4	0.442107	0.617076	0.358626	2.466947	4.979503	0.222886	0.176463	2.
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309_5	0.434940	0.617430	0.358802	2.365785	4.718679	0.213106	0.173627	2.
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5 rows × 81 columns



```
In [4]: data = pd.concat([data[(data["class"] == 'c-CS-s')], data[(data["class"] == 't-CS-s')]]
data["class"].value_counts()
```

```
Out[4]: class
c-CS-s    135
t-CS-s    105
Name: count, dtype: int64
```

There are 135 measurements of mouses of the class c-CS-s and 105 of the class t-CS-s.

b) Create a parallel coordinates plot with plotly from the following 5 proteins: (pPKCG N, pP70S6 N, pS6 N, pGSK3B N, ARC N). Assign different colors to the two selected classes. Annotate every axis with the correct protein name. (9P)

```
In [5]: data.insert(0, "class_id", [(0 if i == 'c-CS-s' else 1) for i in data["class"]])
```

```
In [9]: fig = px.parallel_coordinates(data, color="class_id",
                                     dimensions=['pPKCG_N', 'pP70S6_N', 'pS6_N', 'pGSK3B_N', 'ARC_N'],
                                     color_continuous_scale=px.colors.diverging.Tealros)
fig.show()
```

c) Explore the data by interacting with the parallel coordinates plot. Do you find anything suspicious about the data set? (1P)

The data set looks fairly normal to me. We see the data points evenly distributed around the range of the variables, and even though the subgroups have different distributions, the correlation between variables seems to be the same in all cases(positive, negative, positive, positive). We have some apparent outliers: an individual on group 1 with low values on all proteins, another individual on group 1 that has a particularly high value on protein pS6_N, and 2 individuals on group 0 with high values on proteins pS6_N, pGSK3B_N and ARC_N